

Roll No.

DEEM-08

**B. TECH. (MECHANICAL ENGINEERING)
(SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023
SUPPLY CHAIN MANAGEMENT**

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What is management ? Also define the 'POSDCORB' in detail.

(CO1/CO2)

OR

- (b) What is supply chain network ? Write the key issues in supply chain management.

(CO1/CO2)

2. (a) What are the factors responsible for making supply chain strategies ?
What is bullwhip effect ? How does it affect the supply chain management ?

(CO2)

OR

- (b) What is logistics network ? Also write a short note on transport strategy.

(CO2)

P. T. O.

(2)

3. (a) Explain the principles of 'plant layout' and functional aspects of a factory building. Also give the impact of globalization of location of organizations. (CO2, CO3)

OR

- (b) Write the various methods for selecting plant location. Describe the location factor rating. (CO2, CO3)

4. (a) What is supply chain management ? What are the drivers of supply chain performance ? (CO2, CO3)

OR

- (b) What is a vehicle routing and scheduling method ? (CO2, CO3)

5. (a) What is inventory ? What are components of costs that are affected by inventory decisions ? (CO2, CO3)

OR

- (b) What is the difference between multimodal and unimodal ? What are *four* types of intermodal transportation ? (CO2, CO3)

Roll No.

DEDE-09

**B. TECH. (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023**

FINITE ELEMENT METHOD

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Give *five* applications of finite element method. (CO1)

OR

- (b) What are the advantages of using finite element method ? (CO1)

2. (a) Describe the procedure involved in finite element analysis. (CO1)

OR

- (b) What do you understand by stiffness matrix ? (CO1)

3. (a) Explain what is spring element ? Where can it be used ?

(CO1 & CO2)

OR

- (b) Derive the element matrix equation for spring element with a suitable diagram. (CO1 & CO2)

P. T. O.

(2)

4. (a) How can two spring elements be combined in series ? (CO2)

OR

- (b) What is the difference between a global matrix and element matrix ?
(CO2)

5. (a) What do you understand by homogeneous boundary condition ?
(CO2)

OR

- (b) How can spring elements be combined in parallel ?
(CO2)

Roll No.

DEDE-09

**B. TECH. (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023**

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Maximum Marks : 50

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2. (a) Describe the procedure involved in finite element analysis. (CO1)

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- (b) What do you understand by stiffness matrix ? (CO1)

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(CO1 & CO2)

OR

- (b) Derive the element matrix equation for spring element with a suitable diagram. (CO1 & CO2)

P. T. O.

Roll No.

TAS-701

B. TECH. (MECHANICAL ENGG.) (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

FLIGHT DYNAMICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any of the sub questions.

(ii) Each sub-question carries 10 marks.

(iii) Draw neat diagrams as necessary.

(iv) Make assumptions for any missing data.

1. (a) Explain with a neat sketch the working principle of Pilot-Static Tube. (CO1)

OR

- (b) What do you understand by International Standard Atmosphere ? List the ISA values at Sea level for Pressure, Density and Temperature. (CO1)

2. (a) Explain the formation of downwash and how it contributes in lift dependent drag ? (CO2)

OR

- (b) Define Profile and Parasite Drag in details. (CO2)

P. T. O.

(2)

3. (a) Draw C_l vs. Alpha curve for a symmetrical airfoil with and without flap and explain the curve in detail. (CO1)

OR

- (b) What do you understand by High lift devices ? Explain the following how these devices contribute in further increase in lift. (CO1)
4. (a) Why flaps are lowered during landing and takeoff an airplane. (CO2)

OR

- (b) Define static, dynamic and neutral stability. (CO2)
5. (a) During Stalling of an aircraft, C_l is maximum or minimum. Explain in detail with neat sketch. (CO3)

OR

- (b) What are the design constraints while designing low speed Aircraft ? (CO3)

Roll No.

TME-707

B. TECH. (ME) (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

AUTOMOBILE ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Define Automobile. Give the typical specifications of an automobile.

(CO1)

OR

- (b) Why petrol engines are preferred for two wheelers ?

(CO1)

2. (a) State the advantage of Unibody construction over Body on frame.

(CO2)

OR

- (b) Differentiate between front wheel drive and rear wheel drive. (CO2)

3. (a) Define chassis, frame, body and suspension. What are the different loads acting on chassis ? Explain briefly. (CO3/CO2)

P. T. O.

(2)

OR

(b) What is chassis ? How design is related to vehicle aerodynamics ?

(CO3/CO2)

4. (a) List the forces acting on a chassis frame.

(CO3/CO2)

OR

(b) State the requirements of bodies for various types of vehicles.

(CO3/CO2)

5. (a) Name the different kinds of resistances to vehicle motion. (CO1/CO2)

OR

(b) Explain with suitable reasons for usage of front engine rear wheel drive for most of the Vehicle layout.

(CO1/CO2)

Roll No.

TME-708

**B. TECH. (ME) (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023**

MECHATRONICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Create the block diagram for the three axis vertical CNC milling centre showing all the system constituents like MCU, various drives, ATC, coolant control, automatic door opening/closing system and CNC interface. The block diagram should clearly demonstrate the connectivity of the various CNC elements and feedback control system.

(CO1)

OR

- (b) Draw the block diagram for the fully automatic microprocessor controlled washing machine system after enlisting the various operations that the machine would undertake. Prepare the list of various types of sensors, electrical and electronics devices, interface and hardware devices required in such a system to function properly. (CO1)

P. T. O.

2. (a) By using Laplace transform, determine the ratio $C(s)/R(s)$ for the differential equation given below : (CO2)

$$\frac{d^2c(t)}{dt^2} + 4\frac{dc(t)}{dt} + c(t) = r(t)$$

Given :

$$c(0+) = 0$$

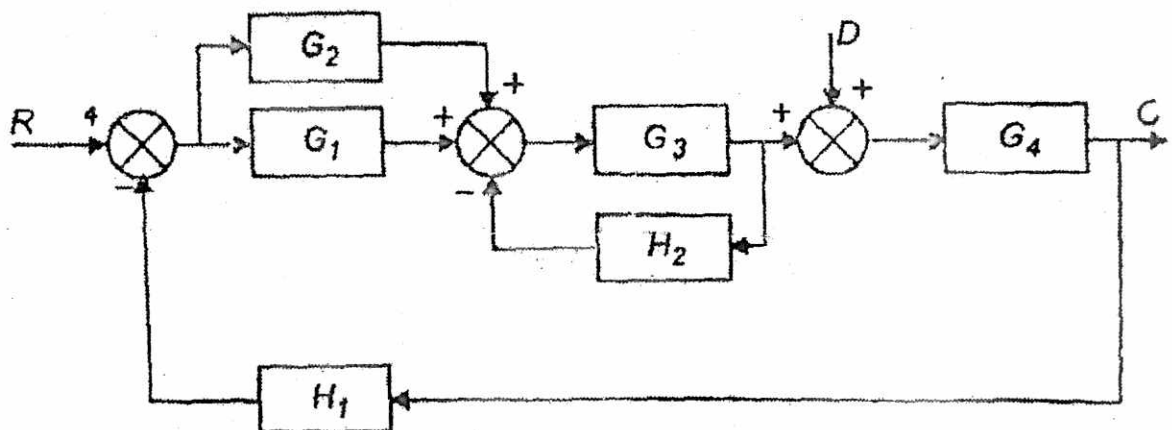
$$c'(0+) = 0$$

Now, if $r(t) = 1$, evaluate time solution for $c(t)$.

OR

- (b) The Laplace transform of $f(t)$ is given by $F(s) = \frac{4}{s(s+2)}$. Find the final value using final value theorem and verify the result by determining $f(t)$ using inverse Laplace transform. (CO2)

3. (a) Derive an expression for the total output for the system represented by the block diagram in below figure : (CO2)



(4)

TME-708

5. (a) What do you mean by mechatronics ? What is its scope ? Explain the evolution of mechatronics through various stages. (CO1)

OR

- (b) What are the types of control system ? Explain the basic elements of a closed loop system with diagram. (CO1)